The sharp second-order cycle bound through six-edge cuts

Collaborative research working draft

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Abstract

Every connected loopless cubic multigraph of cycle rank r with a cyclic edge cut of size at most six satisfies

$$c(G) \le 2^{r-1} + Kr2^{r/2}$$

for an absolute constant K. The known Horak–Kim family shows that the error has sharp order within this class. For large rank, a cyclic six-cut without a smaller cyclic cut gives the stronger strict inequality $c(G) < 2^{r-1}$. Two large shores are handled by incompatible terminal-pairing supports. For a bounded shore, a $K_{3,3}$ subdivision forces a missing four-terminal forest state. Its integral counting deficit absorbs the internal cycles of the other shore. The unrestricted sharp bound remains open.

Status and scope

All new arguments in this report passed independent mathematical audits by the root and three research agents. Exact computations separately check the state identities, the six-cycle certificate, and the subdivision witnesses. The new theorems are not formalized in Lean. Publication priority has not been established. The earlier five-cut proof and the unrestricted forest machinery are explicit inputs, stated below. This report records the new extension; it does not reproduce those earlier proofs.

Rank means $|E| - |V| + \text{comp}$. Paths and cycles are simple, with parallel edges distinguished. A cyclic cut has a cycle on each shore. A normalized k-pole has k distinct degree-two ports and all other degrees three. An earlier relaxation obstruction is included because the new cap inequalities explain its failure.

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1 Inputs from the earlier research packets

The following reports contain the earlier proofs:

- `small-cut-sharp-bound.pdf`: cuts of size at most five, saturation, and protected terminal-path estimates.
- `terminal-path-stability.pdf`: additive terminal-path stability.
- `polynomial-cycle-error.pdf`: unrestricted marked forest and cycle estimates.

The statements needed below are recorded explicitly.

Theorem 1.1 (Established five-cut bound)**.** *There is an absolute K_5 such that every connected loopless cubic multigraph of rank r with a cyclic cut of size at most five satisfies $c(G) \le 2^{r-1} + K_5 r 2^{r/2}$.*

Theorem 1.2 (Established forest and cycle estimates)**.** *For connected subcubic graphs of unbounded rank, cubic outside a fixed connected marked set, the probability that a uniform even edge set is a forest outside that set is at most $1/2 + o(1)$. Also, uniformly over graphs of rank h, one has $c(G) \le (1/2 + o(1))2^h$. The separate unrestricted packet proves a stronger quantitative estimate.*

Lemma 1.3 (Established saturated two-pole classification)**.** *A saturated normalized two-pole consists of bridgeless saturated pieces joined by bridges in series. Each piece is a digon or a capped permutation ladder. Its permutation is a concatenation of singleton blocks and adjacent transpositions.*

Theorem 1.4 (Established path saturation gap)**.** *For a connected graph of rank h and distinct terminals s, t, either $P_{st} = 2^h$ or $P_{st} \le (15/16)2^h$.*

Theorem 1.5 (Established protected high-density theorem)**.** *Fix the number of ports, a bound on the number of protected vertices, and $\varepsilon > 0$. At sufficiently large rank, terminal-path density at least $1/2 + \varepsilon$ in a normalized pole gives two disjoint cyclic vertex sets. They avoid every port and protected vertex, and have boundary at most four in every cubic completion. The rank threshold depends only on the fixed parameters.*

Theorem 1.6 (Aggregate forest states)**.** *For a normalized k-pole Y of rank h, with fixed k, let F_S count forests with odd-degree set S. Then, uniformly as $h \to \infty$,*

$$2^{-h} \sum_{S \text{ even}} F_S \le 2^{k-2} + o(1).$$

Proof. Attach a fixed trivalent tree with k leaves to the k ports, using one edge per leaf. Mark the entire added tree. The graph has rank $h+k-1$, and all vertices outside the marked set are cubic. Every even port pattern has a unique extension through the tree, determined by parity across its edge cuts. Each port parity class has 2^h members. The forest probability outside the marked set is therefore $2^{-(h+k-1)}\sum_S F_S$. Apply the established marked forest estimate and multiply by 2^{k-1}. □

2 Matching augmentation constraints for a fixed number of ports

The protected small-cut theorem applies after adding any fixed matching between ports. It supplies uniform inequalities for the higher terminal-state sectors.

Let Y be a normalized k-pole of rank h. For a nonempty matching J between distinct ports, let $R_J(Y)$ count the following forests. Their odd-degree vertices are exactly the endpoints of J. Their path components define a pairing β on those endpoints. The union of J and β, treated as two matchings with distinguished edge colors, must be one alternating cycle. For $|J|=1$, the alternating cycle has two parallel colored edges, and R_J is the corresponding terminal-path count.

Write $\rho_J = 2^{-h}R_J(Y)$. If J has two edges, R_J is the sum of the two pairing counts other than J. If J has three edges, R_J counts pairing forests whose pairing shares no edge with J.

Proposition 2.1 (Exact augmentation identity)**.** *Let A be a matching on the ports of Y, and choose an edge $e=ab\in A$. Add all edges of $A\setminus\{e\}$ to Y and call the resulting pole Y_A. The ports a,b remain unsaturated. Then*

$$P_{ab}(Y_A)=\sum_{\substack{J\subseteq A\\ e\in J}} R_J(Y).$$

The rank of Y_A is $h+|A|-1$.

Proof. Separate terminal paths according to the subset $I\subseteq A\setminus\{e\}$ of added edges that they use. Deleting those added edges leaves a forest of vertex-disjoint paths. Its odd-degree vertices are exactly the endpoints of $J=I\cup\{e\}$. Adding back I joins these components into one a–b path precisely when J and its forest pairing form one alternating cycle. This gives a bijection with the forests counted by $R_J(Y)$. The disjoint alternatives over I prove the identity. Adding each edge increases the rank by one, giving the rank formula. □

Theorem 2.2 (Uniform matching augmentation inequalities)**.** *Fix k. Suppose Y lies inside a cubic graph with no cyclic cut of size at most four. Uniformly over every nonempty port matching A and every $e\in A$,*

$$\sum_{\substack{J\subseteq A\\ e\in J}} \rho_J \le 2^{|A|-2}+o(1) \qquad (h\to\infty). \tag{1}$$

For $|A|=1$, the right side is $1/2+o(1)$.

Proof. Apply Proposition 2.1. Protect every endpoint of the $|A|-1$ added edges. If the augmented terminal-path density exceeded one half by a fixed amount along unbounded ranks, Theorem 1.5 would apply. It would produce two disjoint cyclic sets avoiding the protected vertices and all remaining ports. Removing the added edges and restoring the original port incidences leaves their cycles and boundaries unchanged. This would give a cyclic cut of size at most four in the original cubic graph.

Thus

$$P_{ab}(Y_A)\le (1/2+o(1))2^{h+|A|-1}.$$

Divide by 2^h and use the exact identity. There are only finitely many matchings for fixed k, so the error is uniform. □

For six ports, a perfect matching $A = \{ab, ij, kl\}$ gives the inequality

$$a_{ab} + \rho_{\{ab,ij\}} + \rho_{\{ab,kl\}} + \rho_{\{ab,ij,kl\}} \le 2 + o(1).$$

The two middle terms are noncap four-terminal forest sums. The last term is the compatible six-terminal forest mass. These inequalities are proved inputs to the six-cut problem. No six-cut error theorem is asserted here.

3 Six-edge cuts: exact states and a relaxation obstruction

This section records exact identities and a finite obstruction to one possible dual argument. It does not prove a six-cut error theorem.

Let normalized six-poles X, Y have ranks h, k. Joining corresponding ports gives rank $r = h + k + 5$, with one-half threshold $16 \cdot 2^{h+k}$. There are fifteen pair-path sectors, fifteen four-port forest sectors, and one six-port forest sector. Let $\mathcal{M}_6$ be the fifteen perfect matchings of the six ports. Write t_α, z_α for the two shores' six-terminal forest counts with pairing $\alpha \in \mathcal{M}_6$. Use P_e, Q_e and $f_{S,\alpha}, g_{S,\alpha}$ for the other states. Put $F_S = \sum_\alpha f_{S,\alpha}$ and $J_S = \sum_\alpha g_{S,\alpha}$.

The exact gluing identity is

$$c(G) = c(X) + c(Y) + \sum_{|e|=2} P_e Q_e + \sum_{|S|=4} \left(F_S J_S - \sum_\alpha f_{S,\alpha} g_{S,\alpha} \right) + \sum_{\substack{\alpha,\beta \in \mathcal{M}_6 \\ \alpha \cap \beta = \varnothing}} t_\alpha z_\beta.$$

Indeed, a cycle crosses the cut in two, four, or six edges. In the six-edge case, the two colored perfect matchings must form one alternating six-cycle. For six ports, this is equivalent to having no common matching edge. Each matching has eight compatible matchings.

Normalize the states of Y by 2^k and call them $a_e, u_{S,\alpha}, w_\beta$. Put $U_S = \sum_\alpha u_{S,\alpha}$. For a pairing α of S, define

$$B_{S,\alpha} = U_S - u_{S,\alpha}.$$

For $\alpha \in \mathcal{M}_6$, put

$$R_\alpha = \sum_{\beta:\, \alpha \cap \beta = \varnothing} w_\beta.$$

The general matching-augmentation theorem gives, in a large pole inside a cubic graph without a cyclic cut of size at most four,

$$a_e \le \tfrac{1}{2} + o(1), \qquad a_e + B_{e \cup f, e|f} \le 1 + o(1)$$

for disjoint pairs e, f. For each perfect matching $\alpha = \{e, f, g\}$ it also gives

$$a_e + B_{e \cup f, e|f} + B_{e \cup g, e|g} + R_\alpha \le 2 + o(1).$$

There are ninety oriented one-edge constraints and forty-five two-edge constraints.

Bounding every opposite-shore state only by its affine-space size gives the universal relaxation

$$\mathcal{L}(Y) = \sum_e a_e + \sum_{|S|=4} \max_\alpha B_{S,\alpha} + \max_{\alpha \in \mathcal{M}_6} R_\alpha.$$

Unlike the five-port case, these constraints do not force a strict deficit below sixteen. The following exact half-integral vector proves this limitation. It is an abstract state vector; no graph realization is asserted.

Label the ports $0, 1, 2, 3, 4, 5$. Set $a_e = 1/2$ on

$$04, 05, 12, 13, 15, 34, 45,$$

and zero on the other pairs. For an increasing four-set $S = \{a, b, c, d\}$, order its pairings as $ab|cd, ac|bd, ad|bc$. The following table gives twice their masses:

S	$2u_S$	S	$2u_S$	S	$2u_S$
0123	0 1 1	0124	0 0 2	0125	0 1 1
0134	1 0 1	0135	1 0 0	0145	0 0 1
0234	1 0 0	0235	0 1 1	0245	0 1 0
0345	1 0 0	1234	1 1 0	1235	0 0 1
1245	1 0 1	1345	1 0 1	2345	1 0 1

Set $w_{02|14|35} = 1/2$ and all other six-terminal masses to zero. Every coordinate is nonnegative, each forest total is at most one, and

$$\sum_e a_e = \frac{7}{2}, \qquad \sum_S U_S = 12, \qquad \sum_\beta w_\beta = \frac{1}{2}.$$

All ninety one-edge inequalities hold with right side one. All forty-five two-edge inequalities hold with right side two. The total-state aggregate bound holds with equality sixteen. Every four-sector table row has a zero entry, and $02|14|35$ is compatible with $01|23|45$. Consequently $\mathcal{L}(Y) = 16$. Thus these inequalities alone cannot yield a universal strict-deficit proof.

The exact checker `check_six_cut_states.py` verifies all these rational inequalities. It also verifies 960 complete gluings and 180 two-edge augmentation identities on four small shores, including a shore with a bridge. The results are in `six-cut-state-checks.json`.

The actual six-cycle shore has more restrictive coefficients than the universal relaxation. Its four-sector vectors are $(1/2, 1/2, 0)$ in cyclic order. Its six-sector mass is split equally between the two alternating perfect matchings of the cycle. An exact rational dual gives crossing maximum at most 15.5 using only matching-augmentation constraints. Section 5 records the certificate and the resulting six-cycle-shore theorem. This additional conclusion does not extend to arbitrary opposite-shore state vectors.

4 Protected cap constraints beyond added matchings

Matching augmentation is one special case of a fixed cap construction. Branching tree caps give additional constraints on six-pole forest states. These constraints exclude the earlier abstract half-integral obstruction. They do not yet prove a bound for every bounded six-cut shore.

Theorem 4.1 (A fixed cap cannot produce high terminal-path density)**.** *Fix an integer $k \geq 2$ and a finite loopless cap graph F. Its vertices have the following types:*

1. *q distinct attachment leaves of degree one.*
2. *p designated terminal vertices of degree two.*
3. *Vertices of degree three.*

Every component of F contains an attachment leaf. Write $g = \beta(F)$ and $c = \mathrm{comp}(F)$. Assume $q \le k$ and $k - q + p \ge 2$.

Let Y be a normalized k-pole of rank h inside a cubic graph with no cyclic cut of size at most four. Identify the attachment leaves of F with any q distinct ports of Y. Call the resulting graph Y_F. It is a normalized $(k - q + p)$-pole of rank

$$\beta(Y_F) = h + d_F, \qquad d_F = q + g - c.$$

For every pair a, b of its ports,

$$2^{-h} P_{ab}(Y_F) \le 2^{d_F - 1} + o(1) \qquad (h \to \infty).$$

For a fixed finite collection of caps, the error is uniform over all caps and port choices.

Proof. The identifications make every filled port cubic. All other degrees have the required values, and each cap component attaches to the connected graph Y. Hence Y_F is connected and normalized. The rank increment is

$$|E(F)| - (|V(F)| - q) = g - c + q = d_F.$$

Protect all new cap vertices and all filled ports of Y. Their total number is fixed, independently of h. If the normalized terminal-path density of Y_F exceeded one half by a fixed amount along unbounded ranks, Theorem 1.5 would apply. It would give two disjoint cyclic vertex sets avoiding every protected vertex and every remaining port. Both sets lie in Y and avoid all of its original ports. Their induced graphs and boundaries are unchanged when the cap is removed and the original completion is restored. Either set would therefore give a cyclic cut of size at most four in the original graph. This contradiction proves the density bound. Finiteness gives the stated uniformity. □

The path count in this theorem is a finite linear combination of the terminal forest states of Y. Here is a precise coefficient description, including caps whose new vertices are the chosen terminals. Let A be the filled ports, and split the chosen pair into $T_Y \subseteq V(Y)$ and $T_F \subseteq V(F) \setminus A$. For $S \subseteq A$, a path restricts to forests in Y, F with odd sets $T_Y \cup S$ and $T_F \cup S$, respectively. Their path pairings must join into one path between the chosen terminals when the vertices of S are identified. For each forest pairing in Y, its coefficient is the number of cap forests with this compatibility property. The empty restriction is included when its odd set is empty. These coefficients depend only on the fixed cap and terminal choices.

The description is exact. A simple path restricts to two forests, since neither restriction can contain a cycle. Conversely, compatible forests have maximum degree two after gluing, exactly two odd vertices, and one nonempty component. Their union is therefore one simple terminal path.

Corollary 4.2 (A three-port star cap)**.** *Let Y be a normalized six-pole of rank h inside a cubic graph with no cyclic cut of size at most four. Choose three ports Q and a pair $e = ab$ among the other three. For disjoint pairs e, f, let $B_{e|f}$ be the normalized sum of the two forest pairings other than $e|f$. Then*

$$a_e + \sum_{f \in \binom{Q}{2}} B_{e|f} \le 2 + o(1).$$

There are sixty such inequalities.

Proof. Attach one new cubic vertex to the three ports of Q. The resulting three-pole has rank $h + 2$. A terminal path either avoids the new vertex or uses exactly two of its incident edges.

Deleting those edges and the new vertex gives one of the two noncap forest pairings on $e \cup f$. Every such forest conversely gives a terminal path. Thus the displayed left side is exactly 2^{-h} times the augmented path count. Apply Theorem 4.1. □

Corollary 4.3 (A four-leaf binary tree cap)**.** *In the same setting, retain terminals $e = ab$ and partition the other four ports as $f \mid g$. Let $R_{e|f|g}$ be the normalized six-terminal forest mass whose pairing has no edge in common with $\{e, f, g\}$. Then*

$$a_e + \sum_{f' \in \binom{[6]\setminus e}{2}} B_{e|f'} + R_{e|f|g} \le 4 + o(1).$$

There are forty-five such inequalities.

Proof. Add two adjacent cubic vertices, attaching one to the two ports of f and the other to the two ports of g. The resulting two-pole has rank $h + 3$. For every two-port boundary pattern in the cap, its unique joining path gives one of the six terms $B_{e|f'}$. For the four-port pattern, the cap's central edge is absent by parity. Its two disjoint paths have pairing $f|g$. They form one terminal path with a forest in Y exactly for the six-terminal pairings counted by $R_{e|f|g}$. The empty cap contributes a_e. Theorem 4.1 now proves the inequality. □

These inequalities exclude the saved half-integral vector in Section 3 as a limit of large poles in this setting. For example, choose terminal pair 14 and star attachment set $\{0, 2, 3\}$. That vector gives

$$a_{14} + B_{14|02} + B_{14|03} + B_{14|23} = 3,$$

whereas Corollary 4.2 gives an upper limit of two. The exact checker finds seven violated star inequalities and seven violated binary-tree inequalities for that vector. This conclusion concerns the saved vector only. It does not show that the enlarged relaxation has a strict universal deficit.

5 A strict deficit for a six-cycle shore

The actual six-cycle state vector admits a stronger estimate than the arbitrary-shore relaxation. Its exact dual uses only matching-augmentation inequalities.

Label the six ports $0, 1, \ldots, 5$ in cyclic order. For a four-set $S = \{a < b < c < d\}$, call $ab|cd$ and $ad|bc$ its noncrossing pairings. Put

$$B_{e|f} = U_{e\cup f} - u_{e|f}$$

for disjoint pairs e, f, as in Section 3. Write R_P for the six-terminal forest mass compatible with a perfect matching P. The two alternating matchings of the six-cycle are

$$P = 01|23|45, \qquad Q = 05|12|34.$$

Lemma 5.1 (The six-cycle functional)**.** *Suppose the normalized states of a six-pole satisfy*

$$a_e \le \tfrac{1}{2} + \eta_1, \qquad a_e + B_{e|f} \le 1 + \eta_2,$$

and

$$a_e + B_{e|f} + B_{e|g} + R_{e|f|g} \le 2 + \eta_3$$

for disjoint pairs in the respective expressions, where $\eta_i \ge 0$. Then its normalized crossing functional against a six-cycle satisfies

$$L_6 \le \frac{31}{2} + \eta_1 + 13\eta_2 + \eta_3.$$

Proof. The six-cycle has two paths between every pair of ports. Each four-sector has one forest in each noncrossing pairing and none in the alternating pairing. Its six-sector has one forest with pairing P and one with pairing Q. Since the six-cycle has rank one, the exact normalized crossing functional is

$$L_6 = \sum_e a_e + \frac{1}{2} \sum_{|S|=4} \Big(B_{\alpha_1(S)} + B_{\alpha_2(S)} \Big) + \frac{1}{2}(R_P + R_Q),$$

where $\alpha_1(S), \alpha_2(S)$ are the two noncrossing pairings of S.

Use the two inequalities

$$a_{01} + B_{01|23} + B_{01|45} + R_P \le 2 + \eta_3,$$

$$a_{05} + B_{05|12} + B_{05|34} + R_Q \le 2 + \eta_3.$$

Use one inequality $a_e + B_{e|f} \le 1 + \eta_2$ for each ordered row (e, f) in the following table:

e	f	e	f	e	f	e	f
03	12	01	24	04	12	25	01
34	01	04	13	35	01	05	13
14	05	34	02	23	04	02	35
23	05	02	45	24	05	03	45
12	34	14	23	12	35	15	23
45	12	24	15	13	45	15	34
45	23	25	34				

Finally use $a_{13} \le 1/2 + \eta_1$ and $a_{35} \le 1/2 + \eta_1$.

The thirty left sides sum exactly to $2L_6$. Indeed, each of the fifteen path coordinates occurs twice. Each noncrossing four-sector quantity occurs once, and the two six-sector quantities occur once. The sum of the right sides is

$$2(2 + \eta_3) + 26(1 + \eta_2) + 2(1/2 + \eta_1) = 31 + 2\eta_1 + 26\eta_2 + 2\eta_3.$$

Divide by two to obtain the stated estimate. □

Theorem 5.2 (Six-cycle-shore error bound)**.** *There is an absolute constant K with the following property. Let G be obtained by joining corresponding ports of a six-cycle and a normalized six-pole Y of rank k. Put $r = k + 6$. Then*

$$c(G) \le 2^{r-1} + Kr2^{r/2}.$$

For sufficiently large k, if G has no cyclic cut of size at most five, then $c(G) < 2^{r-1}$.

Proof. If G has a cyclic cut of size at most five, apply Theorem 1.1. Otherwise Theorem 2.2 gives the hypotheses of Lemma 5.1, with all three errors tending to zero. The crossing contribution, divided by 2^{k+1}, is at most $31/2 + o(1)$. The six-cycle contributes one internal cycle. The other shore contributes at most $(1/2 + o(1))2^k$ cycles. Thus

$$\frac{c(G)}{2^{k+1}} \le \frac{31}{2} + \frac{1}{4} + o(1) = \frac{63}{4} + o(1) < 16.$$

Since $2^{r-1} = 16 \cdot 2^{k+1}$, this gives the strict half bound for large k. The bounded ranks are absorbed by increasing K. □

The exact rational certificate is saved in `six-cycle-shore-rational-dual.json`. The script `verify_six_cycle_shore_dual.py` checks all seventy-five coefficient identities without a numerical solver. The independent script `check_path_six_cycle_dual.py` reconstructs the six-cycle states directly from its edge subsets and verifies the same certificate. This result concerns a six-cycle shore; it does not cover arbitrary cyclic six-edge cuts.

6 A strict six-cut deficit when both shores have large rank

The two shores cannot simultaneously realize the critical states of the one-shore relaxation. The obstruction uses only the matching-augmentation inequalities and the total forest-state bound.

Lemma 6.1 (A matching disjoint from two prescribed matchings)**.** *Any two perfect matchings of six vertices have a common disjoint perfect matching. Here disjoint means that the two matchings share no edge.*

Proof. There are fifteen perfect matchings of six vertices. A fixed matching has eight disjoint perfect matchings. To count these, choose the partner of one vertex in four ways, avoiding its prescribed partner. On the remaining four vertices, exactly one forbidden matching edge remains, leaving two choices. Thus two sets of disjoint matchings, each of size eight, must intersect. □

Define a limiting six-port state vector to consist of nonnegative path coordinates a_e, four-terminal pairing coordinates $u_{S,\alpha}$, and six-terminal pairing coordinates w_β. Put

$$U_S = \sum_\alpha u_{S,\alpha}, \qquad W = \sum_\beta w_\beta, \qquad R_A = \sum_{\beta:\, \beta\cap A=\varnothing} w_\beta.$$

Let $\mathcal{P}$ be the set of these vectors satisfying

$$a_e \le \tfrac{1}{2}, \qquad U_S \le 1, \qquad W \le 1, \qquad \sum_e a_e + \sum_S U_S + W \le 16,$$

and, for each perfect matching $A = \{e, f, g\}$,

$$a_e + (U_{e\cup f} - u_{e|f}) + (U_{e\cup g} - u_{e|g}) + R_A \le 2. \tag{2}$$

The same inequality is required with each of e, f, g as the distinguished edge. Including the one-added-edge inequalities only reduces $\mathcal{P}$; the proof below does not need them.

For two state vectors x, y, let $\mathcal{B}(x, y)$ be their normalized crossing expression from the exact six-cut gluing identity.

Proposition 6.2 (An absolute gap for two limiting state vectors)**.** *There is an absolute constant $\delta_6 > 0$ such that*

$$\mathcal{B}(x, y) \le 16 - \delta_6 \qquad (x, y \in \mathcal{P}).$$

Proof. For one state vector, define

$$E(x) = \sum_e a_e^2 + \sum_S U_S^2 + W^2.$$

Dropping all incompatible-pairing subtractions and using $2uv \le u^2 + v^2$ gives

$$\mathcal{B}(x, y) \le \tfrac{1}{2}(E(x) + E(y)).$$

Since $a_e^2 \le a_e/2$ and the forest totals lie in $[0,1]$,

$$E(x) \le 16 - \tfrac{1}{2}\sum_e a_e \le 16,$$

and the same holds for y.

Suppose equality $\mathcal{B}(x,y) = 16$ holds. Both path sums must vanish. The aggregate bounds then force every one of the fifteen four-terminal totals, and the six-terminal total, to equal one on both shores. Indeed, sixteen numbers in $[0,1]$ must have squared sum sixteen. Equality in the dropped four-sector terms also gives

$$u^x_{S,\alpha} u^y_{S,\alpha} = 0 \qquad \text{for every } S, \alpha. \tag{3}$$

Choose a six-terminal matching β with $w^x_\beta > 0$, and a matching γ with $w^y_\gamma > 0$. They exist because both six-terminal totals equal one. Lemma 6.1 supplies a matching $A = \{e, f, g\}$ disjoint from both. Hence $R^x_A > 0$ and $R^y_A > 0$.

At the proposed equality state, (2) becomes

$$R_A \le u_{e|f} + u_{e|g}, \qquad R_A \le u_{e|f} + u_{f|g}, \qquad R_A \le u_{e|g} + u_{f|g}.$$

If $R_A > 0$, at least two of these three pairing coordinates are positive. Applying this to both shores requires at least four positive coordinates among three shared slots. This contradicts (3). Thus equality is impossible.

The set $\mathcal{P}$ is compact: it is closed, and every coordinate lies in $[0,1]$. The bilinear function $\mathcal{B}$ therefore attains its maximum on $\mathcal{P} \times \mathcal{P}$. Its maximum is strictly below sixteen, proving the proposition. □

Theorem 6.3 (Two large shores of a cyclic six-cut)**.** *There is an absolute integer H with the following property. Let G be a connected loopless cubic multigraph with no cyclic cut of size at most five. Suppose it has a cyclic six-edge cut whose two shores have ranks $h, k \ge H$. Then, with $r = h + k + 5$,*

$$c(G) < 2^{r-1}.$$

Proof. Both shores are connected. A cyclic component of either shore must have at least six boundary edges, since its complement contains a cycle. It therefore uses the entire six-edge cut. Any further component would have no boundary edge, contradicting the connectedness of G.

The cut has distinct endpoints on each shore. If one vertex has at least two cut incidences, its internal degree is at most one. Moving it across the cut preserves cycles on both sides and reduces the cut size to at most five. This would produce a forbidden smaller cyclic cut. Thus the shores are normalized six-poles.

Their actual normalized state vectors satisfy the matching-augmentation inequalities with errors tending to zero as their ranks tend to infinity. They also satisfy the aggregate bound

$$\sum_e a_e + \sum_S U_S + W \le 16 + o(1)$$

from Theorem 1.6. The individual forest totals are at most one exactly. Every convergent subsequence of these state vectors therefore has its limit in $\mathcal{P}$.

Proposition 6.2 and compactness imply that, for sufficiently large h, k, the normalized crossing contribution is at most $16 - \delta_6/2$. Otherwise, a sequence violating this bound would have limiting state vectors contradicting the proposition.

Each shore has at most its affine-space size minus one internal cycles. Their contribution divided by 2^{h+k} is at most $2^{-h} + 2^{-k}$. Increase H until this is less than $\delta_6/2$. The total cycle count is then strictly below $16\,2^{h+k} = 2^{r-1}$. □

This theorem leaves bounded-rank six-cut shores unresolved. The constant gap is established by compactness; no numerical value of δ_6 is asserted here.

7 An explicit gap in the two-shore six-cut relaxation

Retain the limiting state polytope $\mathcal{P}$ and crossing form $\mathcal{B}$ from Section 6. The equality obstruction has a quantitative version.

Lemma 7.1 (Two triples with large pair sums)**.** *Let $x_1, x_2, x_3, y_1, y_2, y_3$ be nonnegative. Suppose every pair sum among the x_i is at least $p \geq 0$, and every pair sum among the y_i is at least $q \geq 0$. Then*

$$\sum_{i=1}^{3} x_i y_i \geq \frac{3}{4} pq.$$

Proof. The assertion is immediate if $p = 0$. If every $x_i \geq p/2$, then $\sum_i y_i \geq 3q/2$ gives the result. Otherwise relabel so that $x_1 = a < p/2$. The other two coordinates are at least $p - a$. Put $Y = y_2 + y_3$ and $Z = y_1 + y_2 + y_3$. Then $Y \geq q$, $Z \geq 3q/2$, and

$$\sum_i x_i y_i \geq a y_1 + (p - a) Y = \left(1 - \frac{2a}{p}\right) pY + \frac{2a}{p} \frac{p}{2} Z \geq \frac{3}{4} pq.$$

□

Proposition 7.2 (A rational six-cut gap)**.** *For every $x, y \in \mathcal{P}$,*

$$\mathcal{B}(x, y) < 16 - \frac{1}{40}.$$

Proof. Suppose $\mathcal{B}(x, y) \geq 16 - d$, where $d = 1/40$. The energy bounds give $E(x), E(y) \leq 16$ and $\mathcal{B}(x, y) \leq (E(x) + E(y))/2$. Thus each energy is at least $16 - 2d$.

For one shore, put

$$D = 16 - \sum_S U_S - W.$$

Its aggregate bound gives $\sum_e a_e \leq D$. Hence

$$E \leq \tfrac{1}{2} \sum_e a_e + \sum_S U_S + W \leq 16 - D/2,$$

so $D \leq 4d$ on either shore. In particular, $W_x, W_y \geq 1 - 4d$.

Let

$$O = \sum_{S,\alpha} u^x_{S,\alpha} u^y_{S,\alpha}$$

be the four-sector pairing overlap. Dropping all incompatible six-pairing products bounds $\mathcal{B}(x, y) + O$ by $(E(x) + E(y))/2 \leq 16$. Therefore $O \leq d$.

For a perfect matching $A = \{e, f, g\}$, write

$$\Delta_A = (1 - U_{e \cup f}) + (1 - U_{e \cup g}) + (1 - U_{f \cup g}).$$

The three augmented inequalities imply that the three pair sums among $u_{e|f}, u_{e|g}, u_{f|g}$ are at least

$$p_A = (R_A - \Delta_A)_+.$$

Use superscripts x, y for the two shores. Every four-sector pairing slot extends uniquely to a perfect matching of six ports. Lemma 7.1 therefore gives

$$O \geq \frac{3}{4} \sum_A p_A^x p_A^y.$$

For nonnegative R, S, D, E, one has $(R-D)_+(S-E)_+ \geq RS - DS - ER$. Also $R_A^x, R_A^y \leq 1$ and

$$\sum_A \Delta_A = 3 \sum_S (1 - U_S) \leq 3D \leq 12d.$$

Consequently,

$$O \geq \frac{3}{4} \left(\sum_A R_A^x R_A^y - 24d \right).$$

Any two six-matchings have at least one common disjoint matching, by Lemma 6.1. Expanding the last sum therefore gives

$$\sum_A R_A^x R_A^y \geq W_x W_y \geq (1 - 4d)^2.$$

Combining the bounds yields

$$O \geq \frac{3}{4}((1-4d)^2 - 24d) = \frac{63}{400} > \frac{1}{40} = d.$$

This contradicts $O \leq d$ and proves the proposition. □

The constant $1/40$ is deliberately conservative. The proof uses no numerical optimization and no graph-realizability assumption beyond membership in the stated limiting polytope.

8 A path gap on every six-pole of girth at least seven

This observation applies at every rank. It strengthens the bounded-shore reduction, but does not by itself prove a six-cut bound.

Lemma 8.1 (Planarity of a saturated pole)**.** *Let X be a connected normalized k-pole, with two ports s, t. If $P_{st}(X) = 2^{\beta(X)}$, then X is planar.*

Proof. Suppress all degree-two ports except s, t. A loop cannot arise: it would be an off-route cycle, which makes the normalized terminal-path count at most one half. The result is a saturated normalized two-pole of the same rank. Apply Lemma 1.3.

Each bridgeless piece is a digon or a capped permutation ladder. The permitted ladder blocks are singletons and adjacent transpositions. Complete each block by adding an edge between its two left attachments, and another between its two right attachments. For a singleton, the completed multigraph has two vertices and three parallel edges. For a transposition, it is K_4. Both are planar.

A planar two-edge join preserves planarity. Indeed, put one designated edge of each planar graph on its outer face, remove these edges, and join corresponding endpoints across the two disks. Apply this operation to successive completed blocks. At the two ends, subdivide the remaining completion edges to form the terminal caps. This constructs the capped ladder as a planar graph. Joining planar pieces by bridges also preserves planarity. Finally, restore the suppressed ports by subdivision. □

Proposition 8.2 (Uniform path gap in a six-pole of large girth)**.** *Let X be a connected normalized six-pole of girth at least seven and rank h. Every terminal pair satisfies*

$$P_{st}(X) \le \frac{15}{16}\, 2^h.$$

Proof. If some pair were saturated, Lemma 8.1 would make X planar. Let n, e, f denote its numbers of vertices, edges, and faces in a planar embedding. The degree sum gives $2e = 3n - 6$. The graph has minimum degree two and contains a cycle. Every facial boundary walk therefore has length at least seven, even when bridges occur. Thus $2e \ge 7f = 7(2-n+e)$, or $5e \le 7n-14$. Combining the two inequalities gives $n \le 2$. This contradicts the six distinct ports. The terminal-path saturation gap now proves the assertion. □

The facial-walk assertion uses minimum degree two. A facial walk with no cycle in its support would force a leaf in its finite tree support. At such a leaf, the facial walk turns back immediately, so the original graph has degree one there. This is impossible. Equivalently, apply the usual planar girth inequality after deleting bridges and summing over the cyclic blocks.

Remark 8.3 (Scope)**.** *The argument does not claim that every planar six-pole is saturated. It only excludes saturation when the girth is at least seven. The preceding six-cycle-shore theorem already handles the smaller-girth cores relevant to the cycle-count problem.*

9 Full four-terminal forest sectors force planarity

The distinction between arbitrary terminal-state vectors and graph states is essential for six-edge cuts. The next theorem supplies a geometric obstruction to simultaneous saturation of all four-terminal forest sectors.

Theorem 9.1 (Planarity from full four-terminal sectors)**.** *Let X be a connected loopless graph of maximum degree three. Let $p_1, \dots, p_k$ be distinct degree-two ports, where $k \ge 6$. Suppose every edge set with odd-degree vertex set equal to any four selected ports is a forest. Then X is planar.*

Equivalently, for a connected normalized k-pole of rank h, the hypothesis says that every four-terminal forest total equals 2^h.

Proof. Suppose X is nonplanar. Kuratowski's theorem supplies a subdivision of K_5 or $K_{3,3}$ as a subgraph. The maximum degree excludes a subdivision of K_5. Thus X contains a subdivision K of $K_{3,3}$. Call its nine subdivided edge paths the core paths.

Each branch vertex of K already has three incident edges in K. It cannot be a port or have a neighbor outside K. Every port lying in K is therefore internal to one core path. Assign that port to its core path.

Each component of $X - V(K)$ has an edge to K, since X is connected. Choose one such attachment edge for each component. Its endpoint in K is internal to a core path, by the degree-three condition at branch vertices. Assign every port in that component to the core path containing this chosen attachment. This assigns all k ports to the nine core paths.

For each core edge e, let w_e count the ports assigned to its path. The graph $K_{3,3}$ has nine four-cycles. Each core edge occurs in four of these cycles and is omitted from five. Consequently, the average total assigned weight on edges omitted from a four-cycle is

$$\frac{5}{9}\sum_e w_e = \frac{5k}{9} \ge \frac{10}{3} > 3.$$

Choose a four-cycle whose omitted edges have total assigned weight at least four. Let C be its subdivided cycle in K.

The two branch vertices outside this four-cycle remain joined by their core path after deleting $V(C)$. Each of the other four omitted core paths has one of these two branch vertices as an endpoint. Its surviving interior remains connected to that endpoint. Thus all surviving interiors of the five omitted core paths lie in one component of $K - V(C)$.

Every port assigned to an omitted core path survives in the same component of $X - V(C)$. For a port inside K, this follows from the preceding paragraph. For a port outside K, its entire outside component survives, and its chosen attachment joins it to that surviving core path. Hence one component of $X - V(C)$ contains at least four ports.

Choose four such ports and call their set S. A connected graph admits an edge set with odd-degree vertex set exactly S, since $|S|$ is even. For example, use the unique parity solution on a spanning tree of that component. This edge set is disjoint from C. Adding $E(C)$ gives an edge set in X with odd-degree set exactly S that contains a cycle. This contradicts the stated forest hypothesis. Therefore X is planar. □

Corollary 9.2 (A missing forest state in every six-pole of girth at least seven)**.** *Let X be a normalized six-pole of rank h and girth at least seven. Then at least one four-terminal forest total is at most $2^h - 1$.*

Proof. Suppose every four-terminal forest total equals 2^h. Theorem 9.1 makes X planar. Write $n = |V(X)|$ and $e = |E(X)|$. Its degrees give $2e = 3n - 6$.

The graph has minimum degree two and contains a cycle. In a planar embedding, every face boundary walk has length at least seven. Indeed, deleting bridge excursions from a face boundary leaves at least one cycle, whose length is at least the girth. Euler's formula and the sum of face-boundary lengths yield

$$2e \geq 7(e - n + 2), \qquad 2e \leq \frac{14}{5}(n - 2).$$

Combining this with the degree sum gives $n \leq 2$. This is impossible because the six ports are distinct. Thus some forest total is strictly smaller than 2^h. Its integrality gives the stated bound. □

The proof requires saturation only for four-terminal forest sectors. It does not require saturation of the six-terminal sector. The missing-state gap is exactly at least 2^{-h} after normalization. This gap is useful when the opposite six-cut shore has unbounded rank.

10 The sharp error for every cyclic cut of size at most six

The bounded-shore case needs an exact counting deficit. A missing forest in one parity class supplies that deficit.

Proposition 10.1 (A bounded six-pole shore of girth at least seven)**.** *Fix an integer $H \geq 1$. Let normalized six-poles X, Y have ranks $1 \leq h \leq H$ and k, respectively. Assume that X has girth at least seven. Join corresponding ports to form a cubic graph G of rank $r = h + k + 5$. Then, uniformly over these choices, for all sufficiently large k,*

$$c(G) < 2^{r-1}.$$

Proof. A planar normalized six-pole cannot have girth at least seven, by the Euler argument in Proposition 8.2. Thus X is nonplanar. Theorem 9.1 supplies a four-port set S_0 for which $F_{S_0}(X) < 2^h$. Write $\delta = 2^{-h}$. Integrality gives

$$2^{-h} F_{S_0}(X) \le 1 - \delta.$$

No terminal pair in X is saturated, by Lemma 8.1. Integrality again gives $2^{-h} P_e(X) \le 1 - \delta$ for every terminal pair e. All other normalized forest totals are at most one.

Use a_e, U_S, W for the normalized path, four-terminal, and six-terminal totals of Y. Let

$$A_Y = \sum_e a_e + \sum_S U_S + W, \qquad T_Y = \sum_{S \neq S_0} U_S + W.$$

There are fourteen four-port terms in T_Y, and one six-port term. Consequently $T_Y \le 15$. The fixed-port aggregate theorem gives $A_Y \le 16 + \eta_k$, with $\eta_k \to 0$ uniformly.

Drop the incompatible-pairing subtractions in the exact six-cut crossing expression. The coefficients of all path sectors and of S_0 are at most $1 - \delta$. Every remaining coefficient is at most one. Therefore the normalized crossing count satisfies

$$\mathcal{B}(X, Y) \le (1 - \delta) A_Y + \delta T_Y \le 16 - \delta + (1 - \delta)\eta_k.$$

This estimate uses the exact quantum δ, rather than an unspecified strict inequality.

The asymptotic cycle bound gives $c(Y) \le (1/2 + \theta_k) 2^k$, where $\theta_k \to 0$ uniformly. The trivial bound gives $c(X) \le 2^h - 1$. Thus

$$\frac{c(G)}{2^{h+k}} \le 16 - \frac{\delta}{2} + (1 - \delta)\eta_k + \delta\theta_k + 2^{-k}.$$

Here $\delta \ge 2^{-H}$. Choose k sufficiently large that the three error terms total less than 2^{-H-1}. The displayed quantity is then strictly below sixteen, uniformly in X and $h \le H$. □

Theorem 10.2 (A strict deficit for genuine six-cuts)**.** *There is an absolute integer R_6 with the following property. Let G be a connected loopless cubic multigraph of rank $r \ge R_6$. Assume that G has a cyclic cut of size six and no cyclic cut of size at most five. Then*

$$c(G) < 2^{r-1}.$$

Proof. The cut shores are connected normalized six-poles, as proved in Theorem 6.3. Write their ranks as $h \le k$.

First suppose that G has a cycle of length at most six. For sufficiently large rank, its complement contains a cycle. A shortest such cycle is induced. A cycle of length at most five would therefore give a forbidden smaller cyclic cut. If its length is six, the six-cycle-shore theorem gives the asserted strict deficit for sufficiently large rank. The normalization of the opposite shore follows from the absence of smaller cyclic cuts.

We may therefore assume that G has girth at least seven. Let H be the threshold from Theorem 6.3. If $h, k \ge H$, that theorem proves the claim. Otherwise $h < H$ and $k = r - h - 5$ tends to infinity with r. Proposition 10.1, with the fixed bound H, proves the claim uniformly. Increase R_6 to meet all these thresholds. □

Corollary 10.3 (Sharp order for cuts of size at most six)**.** *There is an absolute constant K such that every connected loopless cubic multigraph of rank r with a cyclic cut of size at most six satisfies*

$$c(G) \le 2^{r-1} + Kr2^{r/2}.$$

The order of the error is necessary within this class, by the Horak–Kim cyclic-square family.

Proof. For a cyclic cut of size at most five, apply the established sharp small-cut theorem. For a six-cut without any smaller cyclic cut, apply Theorem 10.2 when $r \geq R_6$. The finitely many remaining ranks are covered by increasing K, using $c(G) \leq 2^r - 1$. The cited family has cyclic four-cuts and excess of order $r2^{r/2}$. □

The unrestricted sharp bound is still open. Its remaining large cubic kernels may now be assumed to have no cyclic cut of size at most six and to have girth at least seven.

11 Verification and remaining work

The exact six-cut checker verifies 960 complete gluings and 180 two-edge augmentation identities. Two independent solver-free checks verify the six-cycle coefficient certificate. The new subdivision checker tests all 3,003 assignments of six ports to the nine $K_{3,3}$ core paths. It constructs a nonforest four-port parity witness for each assignment and checks 48,048 affine parity states. It also checks 500 outside-attachment fixtures, including components with several attachments. These checks support the finite identities; the asymptotic conclusions follow from the proofs above.

The unrestricted $O(r2^{r/2})$ bound is still open. The established reductions now allow the remaining large cubic kernels to be assumed cyclically seven-edge-connected, with girth at least seven. The strongest unrestricted estimate remains

$$\psi(r) \leq 2^{r-1} + O\left(\frac{2^r (\log_2 r)^{11/2}}{\sqrt{r}}\right).$$

The shared live handoff records subsequent partial work separately.

References

[1] P. Horak and D. Kim, *Connected cubic graphs with the maximum number of perfect matchings*, Journal of Graph Theory 99 (2022), 671–690. Updated author version: https://arxiv.org/abs/2006.13459.